In [2]:

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.impute import SimpleImputer
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, classification_report,confusion_

# Load the Iris dataset
df = pd.read_csv('iris.csv')
df.info()

df.head()
```

```
c:\users\vamsi2001\appdata\local\programs\python\python39\lib\site-packages
py\_distributor_init.py:30: UserWarning: loaded more than 1 DLL from .libs:
c:\users\vamsi2001\appdata\local\programs\python\python39\lib\site-packages
py\.libs\libopenblas.EL2C6PLE4ZYW3ECEVIV3OXXGRN2NRFM2.gfortran-win_amd64.dll
c:\users\vamsi2001\appdata\local\programs\python\python39\lib\site-packages\num
py\.libs\libopenblas64__v0.3.21-gcc_10_3_0.dll
  warnings.warn("loaded more than 1 DLL from .libs:"
c:\users\vamsi2001\appdata\local\programs\python\python39\lib\site-packages\sci
py\__init__.py:146: UserWarning: A NumPy version >=1.16.5 and <1.23.0 is requir
ed for this version of SciPy (detected version 1.24.3
  warnings.warn(f"A NumPy version >={np_minversion} and <{np_maxversion}"

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   sepal.length  150 non-null    float64
 1   sepal.width   150 non-null    float64
 2   petal.length  150 non-null    float64
 3   petal.width   143 non-null    float64
 4   variety       150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

Out[2]:

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	NaN	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa

In [3]:
```
print(df.isnull().sum())
```

```
sepal.length    0
sepal.width     0
petal.length    0
petal.width     7
variety         0
dtype: int64
```

In [4]:
```
# Handling missing values by replacing them with the column mean
imputer = SimpleImputer(strategy='mean')
df[['petal.width']] = imputer.fit_transform(df[['petal.width']])

#print("\nData after handling missing values:\n", df.head(15))

df.isnull().sum()
```

Out[4]:
```
sepal.length    0
sepal.width     0
petal.length    0
petal.width     0
variety         0
dtype: int64
```

In [5]:
```
# Prepare features and labels
X = df.drop(columns=['variety'])  # Features (sepal and petal measurements)
y = df['variety']  # Target variable (species of iris)

# Split the data into training and test sets (80% train, 20% test)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, ran
```

In [6]:
```
# standarise the features to scale values between -1 and 1
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
X_train
```

Out[6]:
```
array([[-1.47393679,  1.20365799, -1.56253475, -1.37183001],
       [-0.13307079,  2.99237573, -1.27600637, -1.0990868 ],
       [ 1.08589829,  0.08570939,  0.38585821, -0.00429936],
       [-1.23014297,  0.75647855, -1.2187007 , -0.00429936],
       [-1.7177306 ,  0.30929911, -1.39061772, -1.37183001],
       [ 0.59831066, -1.25582892,  0.72969227,  0.94648729],
       [ 0.72020757,  0.30929911,  0.44316389,  0.40100087],
       [-0.74255534,  0.98006827, -1.27600637, -1.37183001],
       [-0.98634915,  1.20365799, -1.33331205, -1.37183001],
       [-0.74255534,  2.32160658, -1.27600637, -1.50820162],
       [-0.01117388, -0.80864948,  0.78699794,  0.94648729],
       [ 0.23261993,  0.75647855,  0.44316389,  0.53737247],
       [ 1.08589829,  0.08570939,  0.55777524,  0.40100087],
       [-0.49876152,  1.87442714, -1.39061772, -1.0990868 ],
       [-0.49876152,  1.4272477 , -1.27600637, -1.37183001],
       [-0.37686461, -1.47941864, -0.01528151, -0.28085716],
       [ 0.59831066, -0.58505976,  0.78699794,  0.40100087],
       [ 0.72020757,  0.08570939,  1.01622064,  0.81011568],
       [ 0.96400139, -0.13788033,  0.38585821,  0.26462926],
```

In [7]:
```
# Create and train the KNN model (K=5)
knn = KNeighborsClassifier(n_neighbors=5,metric='euclidean')
knn.fit(X_train, y_train)
```

Out[7]:
```
KNeighborsClassifier(metric='euclidean')
```

In a Jupyter environment, please rerun this cell to show the HTML representation or trust the notebook.
On GitHub, the HTML representation is unable to render, please try loading this page with nbviewer.org.

In [16]:
```
#y_test
```

In [8]:
```
# Predict on the test set
y_pred = knn.predict(X_test)
y_pred
```

Out[8]:
```
array(['Versicolor', 'Setosa', 'Virginica', 'Versicolor', 'Versicolor',
       'Setosa', 'Versicolor', 'Virginica', 'Versicolor', 'Versicolor',
       'Virginica', 'Setosa', 'Setosa', 'Setosa', 'Setosa', 'Versicolor',
       'Virginica', 'Versicolor', 'Versicolor', 'Virginica', 'Setosa',
       'Virginica', 'Setosa', 'Virginica', 'Virginica', 'Versicolor',
       'Virginica', 'Virginica', 'Setosa', 'Setosa'], dtype=object)
```

In [9]:
```
confusion_matrix(y_test,y_pred)
```

Out[9]:
```
array([[10,  0,  0],
       [ 0,  9,  0],
       [ 0,  1, 10]], dtype=int64)
```

In [10]:

```
# Evaluate model performance
accuracy = accuracy_score(y_test, y_pred)
print(f"\nModel Accuracy: {accuracy:.2f}")

```

```
Model Accuracy: 0.97
```

In [11]:

```
# Display detailed classification report
print(classification_report(y_test, y_pred))
```

```
              precision    recall  f1-score   support

      Setosa       1.00      1.00      1.00        10
  Versicolor       0.90      1.00      0.95         9
   Virginica       1.00      0.91      0.95        11

    accuracy                           0.97        30
   macro avg       0.97      0.97      0.97        30
weighted avg       0.97      0.97      0.97        30
```

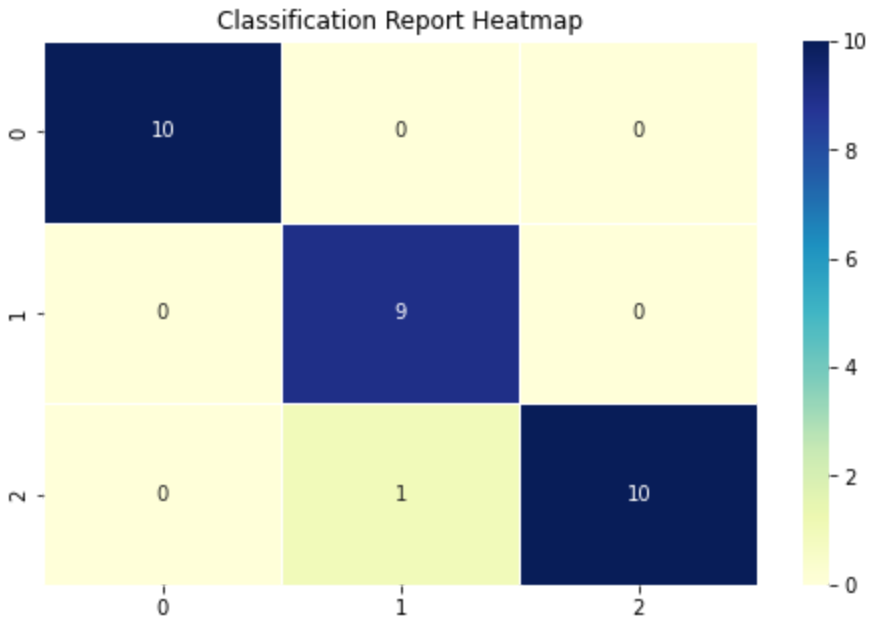
In [17]:

```
import seaborn as sns
import matplotlib.pyplot as plt
# Generate classification report as a dictionary
class_report_dict = classification_report(y_test, y_pred, output_dict=True)

# Convert classification report to DataFrame
class_report_df = pd.DataFrame(class_report_dict).transpose()


```

In [18]:
```
# Generate heatmap for classification report
import matplotlib.pyplot as plt
import seaborn as sns
plt.figure(figsize=(8, 5))
sns.heatmap(confusion_matrix(y_test,y_pred), annot=True, cmap="YlGnBu", line
plt.title("Classification Report Heatmap")
plt.show()
```



In [14]:
```
#predicting using custom input
sample= [[1.2,2.4,1.7,0.3]]
pred = knn.predict(sample)
pred
```

Out[14]:
```
array(['Virginica'], dtype=object)
```

In [ ]:
```

```

In [ ]:
```

```

In [ ]:
```

```

In [ ]:
```

```